Hydrophone analog board

Feedback for Isaac

Note from Eli

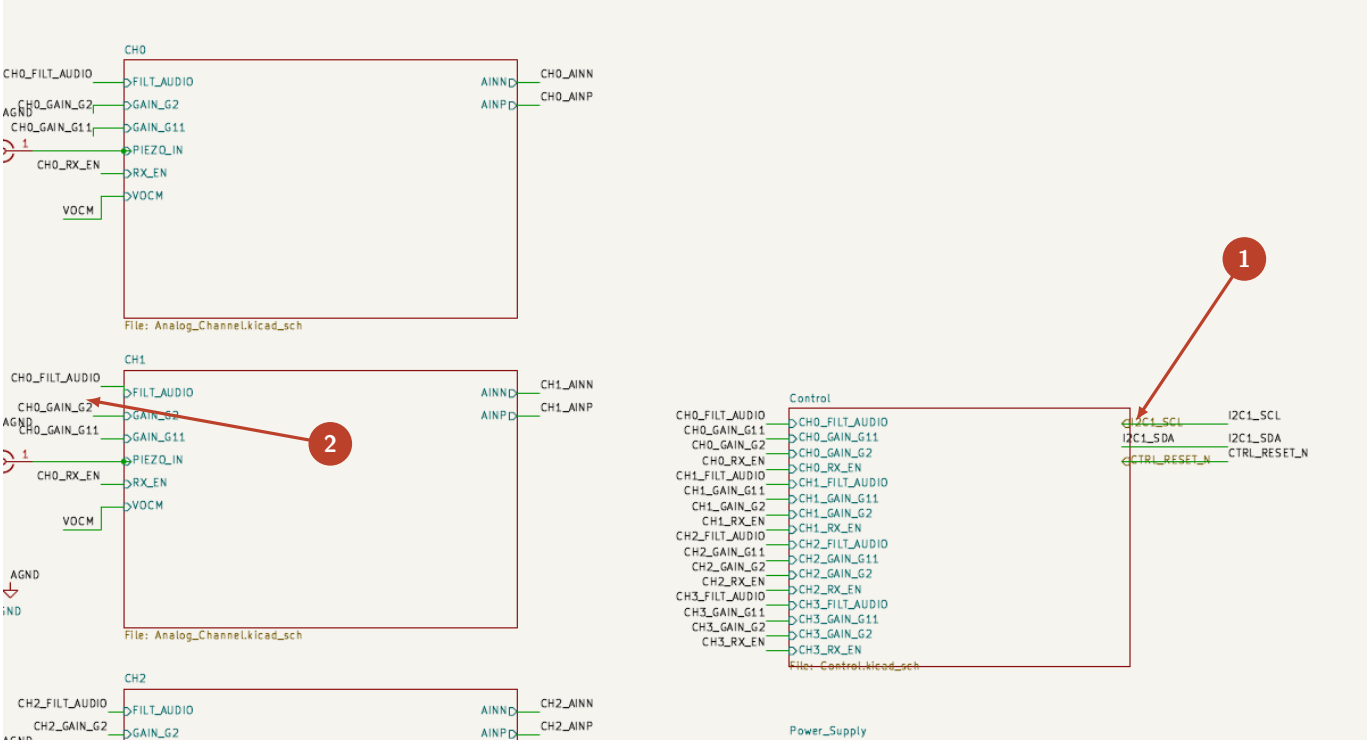
"This is an audit I had my bot performed. I looked over it's results and checked for any mistakes it made and iterated a few times on the feedback. Not saying this is perfect but I think it should catch any of the obvious critical issues, some of which I confirmed manually"

ID	Feedback item	Classification
A1	Connect J5 I^2C/reset through the Control sheet interface	Confirmed defect
A2	Give CH1 its own relay-control labels	Confirmed defect
A3	Connect Q5's gate input to its expander command	Confirmed defect
A4	Provide sufficient input headroom for the 5 V regulator	Confirmed defect
A5	Establish negative-rail load capacity and noise performance	Qualification check
A6	Correct the capacitor pairs selected by the filter relays	Inherited defect

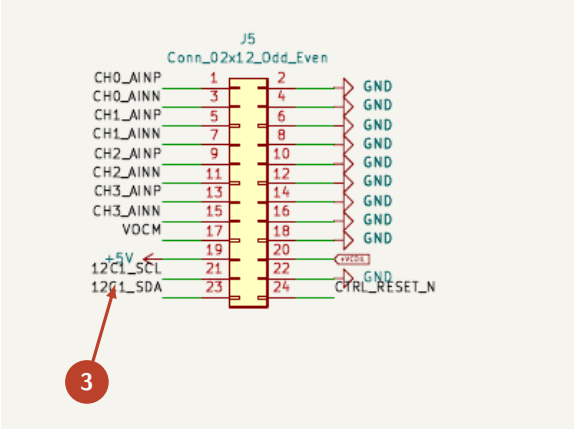
Scope: Receive-only. A6 predates the latest integration changes; A5 requires validation rather than being a confirmed failure.

A1–A2 / Root-sheet connections

Confirmed • Integration wiring



1 — Missing sheet interface. The labels beside Control are separate hierarchical-label objects, not pins on the sheet symbol. The child uses local I2C/reset labels. Thus the header and chips remain on separate nets even after spelling is corrected.

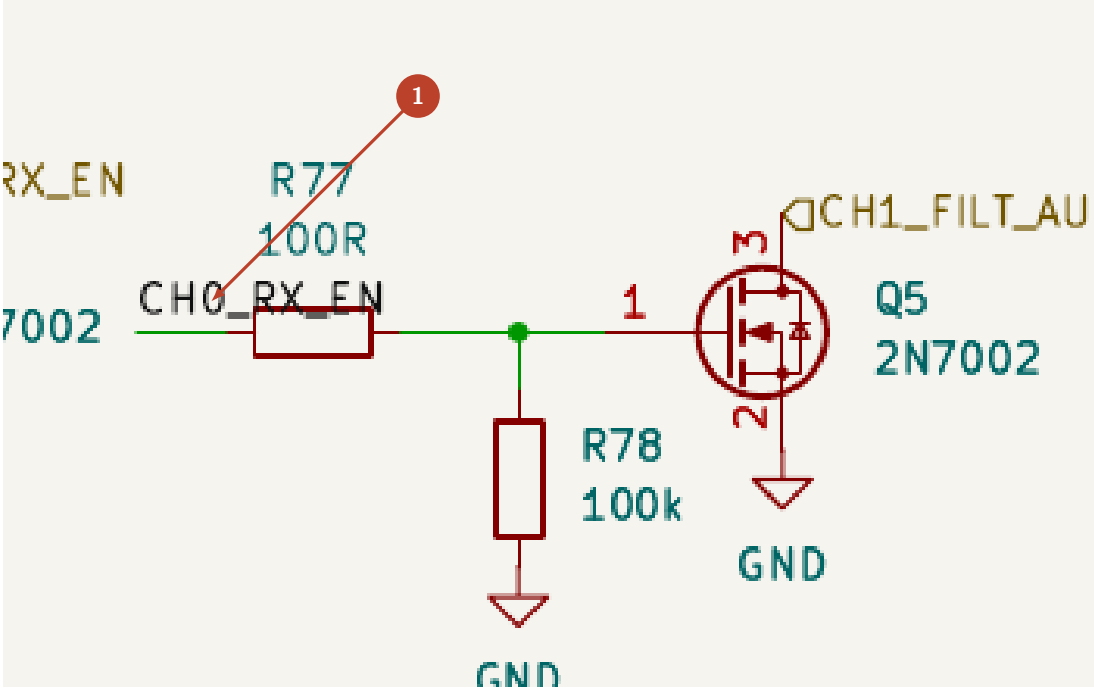


3 — Header spelling. J5.21 and J5.23 use `I2C1_SCL/SDA` (digit 1), not `I2C1_SCL/SDA`. J5.24 reset is also isolated across the sheet boundary.
Check: export the netlist and confirm J5.21 reaches U701–U703.12, J5.23 reaches their pin 13, and J5.24 reaches their pin 18.

2 — CH1 uses CH0 controls. The design has four analog channels, CH0–CH3, and separate driver groups for each. All four control labels entering CH1 repeat CH0; CH2 and CH3 use their own channel labels. For example CH0_RX_EN joins Q4.3 to K1.A1 and K6.A1, while CH1_RX_EN contains Q8.3 alone. Correct the four CH1 labels and verify each channel has its own coil drivers.

A3 / Q5 gate command

Confirmed • Control sheet



1 — R77 is fed by the wrong signal. R77.1 connects to CH0_RX_EN, the low-side relay-coil drive net. R77.2 connects to Q5's gate. The intended command, CTRL_CH1_AUDIO, is present at U701.2 but has no gate load on its net.

Why it matters. This couples channel 1's filter driver to channel 0's receive-coil node rather than to its expander command. Correcting the CH1 root labels alone does not fix this separate error.

Action. Connect R77's input to the intended expander command and inspect the other fifteen MOSFET gate inputs for the same pattern.

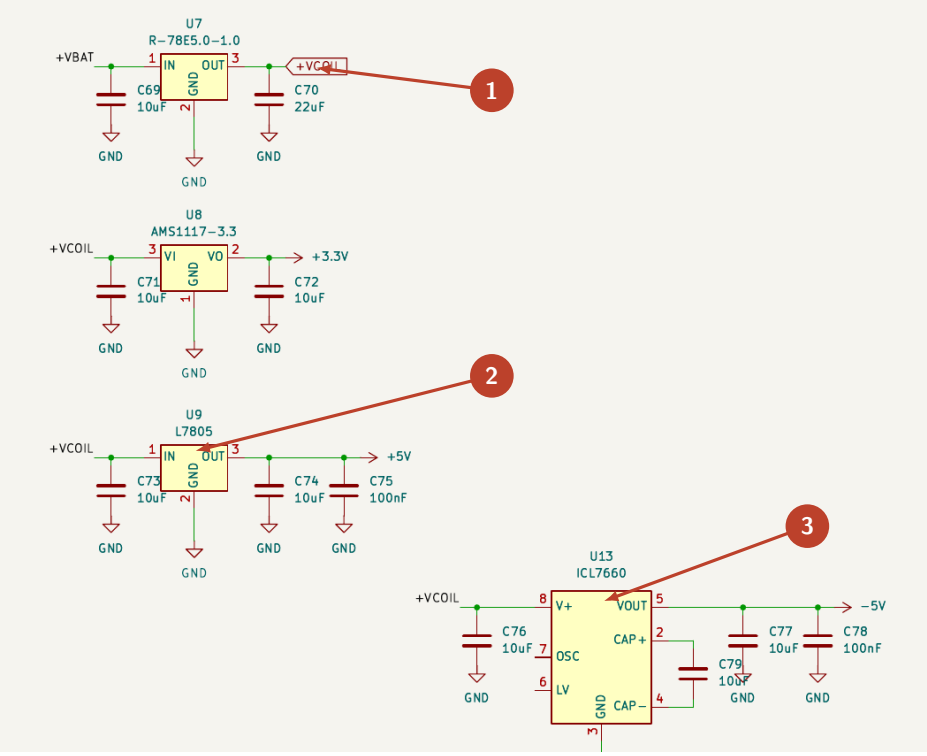
Acceptance checks.

- U701.2 and R77.1 appear on one command net.
- Q5.1, R77.2 and R78.2 remain on the gate net.
- CH0_RX_EN contains only the intended channel 0 coil-drive connections; it no longer feeds R77.
- With corrected root labels, Q5.3 reaches channel 1's filter coils.

Component qualification. Select exact relay ordering codes. Check 5 V coil current, 2N7002 conduction at 3.3 V gate drive, total coil load, flyback release time and analog contact parasitics. Generic relay symbols and assigned footprints do not establish these ratings.

A4–A5 / Supply design

One confirmed headroom error; one major qualification concern



1–2 / A4 — U9 receives 5 V, not the battery voltage. VBAT is the submarine's nominal 16 V battery supply. The drawn path is VBAT → U7 (5 V converter) → +VCOIL → U9 (L7805). U7 has already reduced the voltage to 5 V before it reaches U9. The L7805 needs an input above 5 V to maintain a regulated 5 V output; the present connection does not provide that margin.

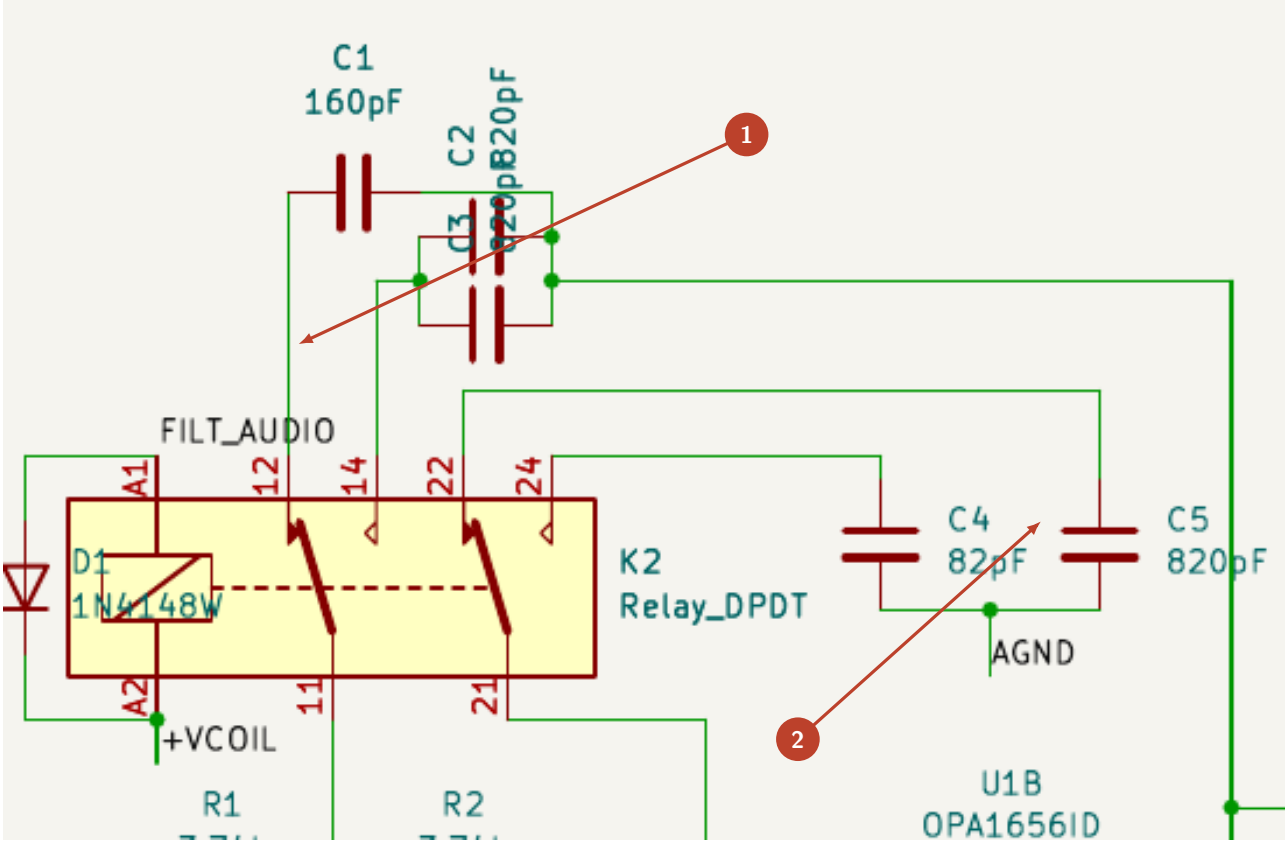
Action. Revise the supply topology. Feeding U9 from VBAT would provide voltage margin, but at nominal 16 V it would dissipate approximately $(16 - 5)I$: 1.1 W per 100 mA. Check the battery's full voltage range, total load, package cooling and noise before selecting that arrangement.

3 / A5 — Negative rail load. Eight OPA1656 dual packages imply about $16 \times 3.9 = 62.4$ mA typical quiescent demand before signal loads. ICL7660 has typical 55 Ω output resistance at the datasheet's 20 mA condition. A credible load/ripple/startup budget is needed, likely with a different supply. 20 mA is *not* asserted as an absolute current limit.

Sources: RECOM R-78E, ST L78, TI OPA1656, Renesas ICL7660.

A6 / Filter capacitor selection

Confirmed • Inherited shared Analog_Channel wiring



1–2 — Opposite capacitor banks selected. On CH0, K2's resting contacts select C1 = 160 pF for feedback but C5 = 820 pF for the shunt. Energized contacts select C2+C3 = 1640 pF and C4 = 82 pF. The desired pairs are 160/82 pF and 1640/820 pF. K5 repeats the mismatch with C12–C16; the shared sheet propagates this to all four channels.

Action. Correct the second-pole capacitor selection in both filter stages. Trace relay common/NC/NO contacts in both states, then validate the cascade response with tolerances and actual relay parasitics.

Impact check. An ideal two-stage unity-gain model gives approximately -3.72 dB at 20 kHz and -35.98 dB at 200 kHz in the resting state, relative to 1 kHz; energized, about $+13.36$ dB at 100 kHz. These are diagnostic calculations, not hardware measurements. Connectivity establishes the mismatch independently of that model.

Attribution. This wiring predates Isaac's latest integration changes. It belongs in the board's correction list, but should not be attributed as a newly introduced mistake in his push.